Severn Lortie

EDUCATION

- **MASc in Robotics** · September 2025–Present
  Smith Engineering, Queen's University, Kingston
  Supervisor: Dr. Jonathan Gammell
  Thesis: *An Autonomous Sailboat and Informative Path Planner for Environmental Monitoring*
- **Bachelor of Computer Engineering** · September 2020–April 2025
  Smith Engineering, Queen's University, Kingston

RESEARCH EXPERIENCE

- **MASc Researcher** · September 2025–Present
  Robotic ESP Lab, Queen's University, Kingston
  Supervisor: Dr. Jonathan Gammell
  - Designed and deployed the complete control, actuation, and motion-planning stack for an autonomous sailboat
  - Validated the system through 6 on-water field tests in Lake Ontario
  - Achieved reliable waypoint following and optimal sail angle control, hitting 8/8 test waypoints in a summer 2026 field test
  - Designed and evaluated a sampling-based informative path planning algorithm that integrated sorted search heuristics for better performance over existing baselines
- **Undergraduate Research Assistant** · May 2025–September 2025
  Robotic ESP Lab, Queen's University, Kingston
  - Programmed a software bridge to ingest data from NMEA 2000, a common protocol for marine sensors, into the Robot Operating System (ROS). This enabled the development of the autonomous sailboat for my Master's research. The code was open-sourced for other researchers to use

WORK EXPERIENCE

- **Software Engineering Intern** · May 2023–August 2024
  Distributive · Kingston, Ontario
  - Built and presented an LLM inference demo on Distributive's parallel-computing platform to Nokia's CTO and AI/HPC lead, contributing to follow-up meetings and a collaboration.
  - Co-developed, from scratch, the company's Python API for its core parallel-computing platform in a two-engineer effort using an embedded JavaScript-in-Python runtime; achieved feature parity with the native JavaScript API and supported several non-trivial workloads.
  - Designed a non-blocking WebSocket bridge in C++ from the embedded V8 workers to the Chrome DevTools Protocol; the resulting debugger was used to fix critical product bugs.

linkedin.com/in/severn-lortie severn.lortie@queensu.ca October 2026

ENGINEERING DESIGN TEAMS AND PROJECTS

- **Ingenuity Labs Racing Team** · RoboRacer/F1Tenth · September 2026–Present
  Planning Lead, Queen's University
  - Participated in three tests of the car and worked to diagnose complex fault, intermittent failure in the motor controller
  - Unified team workflow for track mapping and race line generation, allowing testing to be conducted without large setup overhead
  - Worked with team to diagnose fault that caused car to crash during previous competition, enabling more reliable operation
- **Reducing High Performance Computing Power Consumption with a Proactive CPU Governor** · Capstone Project · September 2024
  Queen's Univeristy
  - Worked with team of three other students to design a custom CPU frequency governor to increase the power efficiency of representative HPC workloads. Testing on a scientific proxy application called Quicksilver, the team was able to achieve a 4.6% decrease in runtime and 7.4% decrease in total energy consumption.
  - My role was to design and program the governor and create the final project poster.

PUBLICATIONS

1. Sanindie Silva, David Courtis, Jax Hodgkinson, and **Severn Lortie**. "Creating a Corn Yield Estimator: A Deep Learning Approach via UAV Imagery"
   Canadian Undergraduate Conference on AI (CUCAI) · 2023

PRESENTATIONS AND POSTERS

- **Severn Lortie**, Meredith Scharlach, and Jonathan Gammell. "The Martin 16 Autonomous Sailboat"
  Robotics and AI Symposium (RAIS) at Ingenuity Labs, Queen's University
  October 2026 · Poster
- Sanindie Silva, David Courtis, Jax Hodgkinson, and **Severn Lortie**. "Creating a Corn Yield Estimator: A Deep Learning Approach via UAV Imagery"
  Canadian Undergraduate Conference on AI (CUCAI) · May 2023 · Oral presentation

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AWARDS AND SCHOLARSHIPS

- **Ontario Graduate Scholarship (OGS)** · $15,000 · 2026-2027
- **Vector Institute Scholarship in AI** · $17,500 · 2025-2026
- **NSERC Undergraduate Student Research Award (USRA)** · $9,000 · May 2025
- **Queen's University Principal's Scholarship** · $4,000 · 2020
- **2nd Place Paper** · Canadian Undergraduate Conference in AI (CUCAI) · $500 · May 2023

- **1st Place Project** · Amazon HACKTO Hackathon · $1,000 · September 2022

TEACHING AND MENTORING EXPERIENCE

- **Teaching Assistant** · Term 20XX
  Course Code: Course Name, University Name
  - Led [tutorials or laboratory sessions] covering [topics].
  - Supported students with [technical concepts, experiments, or assignments].
- **Mentoring Role** · 20XX–20XX
  Program or Organization Name

  Mentored [students or participants] in [subject or activity], helping them [specific responsibility or outcome].

TECHNICAL SKILLS

- **Programming:** [Languages and relevant libraries]
- **Modelling and simulation:** [Software and methods]
- **Experimental methods:** [Measurement, instrumentation, and testing techniques]
- **Design and fabrication:** [CAD software, manufacturing processes, and equipment]

LEADERSHIP AND SERVICE

- **Club or Organization Name** · 20XX–20XX
  Executive or Leadership Role
  - Organized [events, workshops, or initiatives] for [audience].
  - Managed [budget, volunteers, partnerships, or other responsibilities].
- **Outreach or Service Role** · Organization Name · 20XX

  Contributed to [activity] by [specific contribution].